(b) Equation (3.1.17) relates global nodal forces to global nodal displacements as follows:

$$\begin{Bmatrix} F_{1x} \\ F_{2x} \\ F_{3x} \\ F_{4x} \end{Bmatrix} = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} \tag{3.1.18}$$

Invoking the boundary conditions, we have

$$u_1 = 0 \qquad u_4 = 0 \tag{3.1.19}$$

Using the boundary conditions, substituting known applied global forces into Eq. (3.1.18), and partitioning equations 1 and 4 of Eq. (3.1.18), we solve equations 2 and 3 of Eq. (3.1.18) to obtain

$$\begin{Bmatrix} 3000 \\ 0 \end{Bmatrix} = 10^6 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix} \tag{3.1.20}$$

Solving Eq. (3.1.20) simultaneously for the displacements yields

$$u_2 = 0.002 \text{ in.} \quad u_3 = 0.001 \text{ in.} \tag{3.1.21}$$

(c) Back-substituting Eqs. (3.1.19) and (3.1.21) into Eq. (3.1.18), we obtain the global nodal forces, which include the reactions at nodes 1 and 4, as follows:

$$\begin{aligned} F_{1x} &= 10^6(u_1 - u_2) = 10^6(0 - 0.002) = -2000 \text{ lb} \\ F_{2x} &= 10^6(-u_1 + 2u_2 - u_3) = 10^6[0 + 2(0.002) - 0.001] = 3000 \text{ lb} \\ F_{3x} &= 10^6(-u_2 + 2u_3 - u_4) = 10^6[-0.002 + 2(0.001) - 0] = 0 \\ F_{4x} &= 10^6(-u_3 + u_4) = 10^6(-0.001 + 0) = -1000 \text{ lb} \end{aligned} \tag{3.1.22}$$

The results of Eqs. (3.1.22) show that the sum of the reactions F_{1x} and F_{4x} is equal in magnitude but opposite in direction to the applied nodal force of 3000 lb at node 2. Equilibrium of the bar assemblage is thus verified. Furthermore, Eqs. (3.1.22) show that $F_{2x} = 3000$ lb and $F_{3x} = 0$ are merely the applied nodal forces at nodes 2 and 3, respectively, which further enhances the validity of our solution.

3.2 Selecting a Displacement Function in Step 2 of the Derivation of Stiffness Matrix for the One-Dimensional Bar Element

Consider the following guidelines, as they relate to the one-dimensional bar element, when selecting a displacement function. (Further discussion regarding selection of displacement functions and other kinds of approximation functions (such as temperature functions) will be provided in Chapter 4 for the beam element, in Chapter 6 for the constant-strain triangular element, in Chapter 8 for the linear-strain triangular element, in Chapter 9 for the axisymmetric element, in Chapter 10 for the three-noded bar element and the rectangular plane element, in Chapter 11 for the three-dimensional stress element, in Chapter 12 for the plate bending element, and in Chapter 13 for the heat transfer problem. More information is also provided in References [1–3].

Guidelines for Selecting Displacement Functions

1. We must choose in advance the mathematical function to represent the deformed shape of the bar element under loading. Because it is difficult, if not impossible at times, to obtain a closed form or exact solution, we assume a solution shape or distribution of displacement within the element by using an appropriate mathematical function. The most common functions used are polynomials.

Because the bar element resists axial loading only with the local degrees of freedom for the element being displacement u_1 and u_2 along the x direction, we choose a displacement function u to represent the axial displacement throughout the element. Here a linear displacement variation along the x axis of the bar is assumed [Figure 3–4(b)], because a linear function with specified endpoints has a unique path. Therefore,

$$u = a_1 + a_2 x \tag{3.2.1}$$

In general, the total number of coefficients a is equal to the total number of degrees of freedom associated with the element. Here the total number of degrees of freedom is two—an axial displacement at each of the two nodes of the element. In matrix form, Eq. (3.2.1) becomes

$$u = [1 \quad x]\begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix} \tag{3.2.2}$$

We now want to express u as a function of the nodal displacements u_1 and u_2, as this will allow us to apply the physical boundary conditions on nodal displacements directly as indicated in

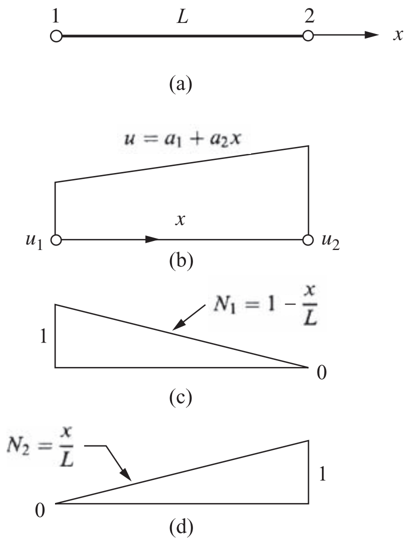

Figure 3–4 (a) Bar element showing plots of (b) displacement function u and shape functions, (c) N_1 and, (d) N_2 over domain of element

step 3 and to then relate the nodal displacements to the nodal forces in step 4. We achieve this by evaluating u at each node and solving for a_1 and a_2 from Eq. (3.2.1) as follows:

$$u(0) = u_1 = a_1 \tag{3.2.3}$$

$$u(L) = u_2 = a_2 L + u_1 \tag{3.2.4}$$

or, solving Eq. (3.2.4) for a_2,

$$a_2 = \frac{u_2 - u_1}{L} \tag{3.2.5}$$

Upon substituting Eqs. (3.2.3) and (3.2.5) into Eq. (3.2.1), we have

$$u = \left(\frac{u_2 - u_1}{L}\right)x + u_1 \tag{3.2.6}$$

In matrix form, we express Eq. (3.2.6) as

$$u = \left[1 - \frac{x}{L} \quad \frac{x}{L}\right]\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \tag{3.2.7}$$

or

$$u = [N_1 \quad N_2]\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \tag{3.2.8}$$

Here

$$N_1 = 1 - \frac{x}{L} \quad \text{and} \quad N_2 = \frac{x}{L} \tag{3.2.9}$$

are called the *shape functions* because the N_i's express the shape of the assumed displacement function over the domain (x coordinate) of the element when the ith element degree of freedom has unit value and all other degrees of freedom are zero. In this case, N_1 and N_2 are linear functions that have the properties that $N_1 = 1$ at node 1 and $N_1 = 0$ at node 2, whereas $N_2 = 1$ at node 2 and $N_2 = 0$ at node 1. See Figure 3–4(c) and (d) for plots of these shape functions over the domain of the spring element. Also, $N_1 + N_2 = 1$ for any axial coordinate along the bar.

The significance of the shape functions summing to one is described more fully under Guideline 4. In addition, the N_i's are often called *interpolation functions* because we are interpolating to find the value of a function between given nodal values. The interpolation function may be different from the actual function except at the endpoints or nodes, where the interpolating function and actual function must be equal to specified nodal values.

2. The approximation function should be continuous within the bar element. The simple linear function for u of Eq. (3.2.1) certainly is continuous within the element. Therefore, the linear function yields continuous values of u within the element and prevents openings, overlaps, and jumps because of the continuous and smooth variation in u (Figure 3–5).

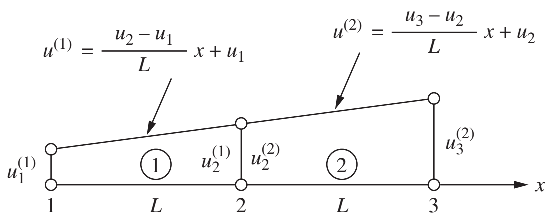

Figure 3–5 Interelement continuity of a two-bar structure

3. The approximating function should provide interelement continuity for all degrees of freedom at each node for discrete line elements and along common boundary lines and surfaces for two- and three-dimensional elements. For the bar element, we must ensure that nodes common to two or more elements remain common to these elements upon deformation and thus prevent overlaps or voids between elements. For example, consider the two-bar structure shown in Figure 3–5. For the two-bar structure, the linear function for u [Eq. (3.2.1)] within each element will ensure that elements 1 and 2 remain connected; the displacement at node 2 for element 1 will equal the displacement at the same node 2 for element 2; that is, $u_2^{(1)} = u_2^{(2)}$. This rule was also illustrated by Eq. (2.3.3). The linear function is then called a *conforming*, or *compatible*, *function* for the bar element because it ensures the satisfaction both of continuity between adjacent elements and of continuity within the element.

In general, the symbol C^m is used to describe the continuity of a piecewise field (such as axial displacement), where the superscript m indicates the degree of derivative that is interelement continuous. A field is then C^0 continuous if the function itself is interelement continuous. For instance, for the field variable being the axial displacement illustrated in Figure 3–5, the displacement is continuous across the common node 2. Hence the displacement field is said to be C^0 continuous. Bar elements, plane elements (see Chapter 7), and solid elements (Chapter 11) are C^0 elements in that they enforce displacement continuity across the common boundaries.

If the function has both its field variable and its first derivative continuous across the common boundary, then the field variable is said to be C^1 continuous. We will later see that the beam (see Chapter 4) and plate (see Chapter 12) elements are C^1 continuous. That is, they enforce both displacement and slope continuity across common boundaries.

4. The approximation function should allow for rigid-body displacement and for a state of constant strain within the element. The one-dimensional displacement function [Eq. (3.2.1)] satisfies these criteria because the a_1 term allows for rigid-body motion (constant motion of the body without straining) and the a_2x term allows for constant strain because $\varepsilon_x = du/dx = a_2$ is a constant. (This state of constant strain in the element can, in fact, occur if elements are chosen small enough.) The simple polynomial Eq. (3.2.1) satisfying this fourth guideline is then said to be *complete* for the bar element.

This idea of completeness also means in general that the lower-order term cannot be omitted in favor of the higher-order term. For the simple linear function, this means a_1 cannot be omitted while keeping a_2x. Completeness of a function is a necessary condition for convergence to the exact answer, for instance, for displacements and stresses (Figure 3–6) (see Reference [3]). Figure 3–6 illustrates monotonic convergence toward an exact solution for displacement as the number of elements in a finite element solution is increased. Monotonic convergence is then the process in which successive approximation solutions (finite element solutions) approach the exact solution consistently without changing sign or direction.